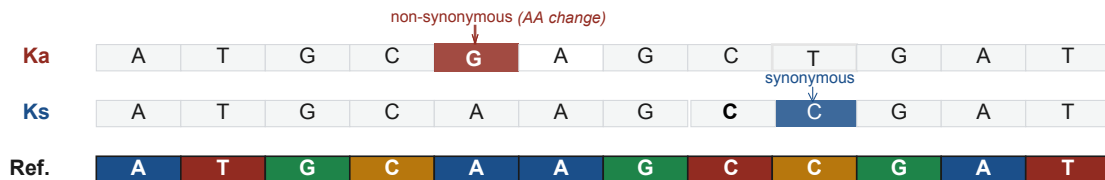


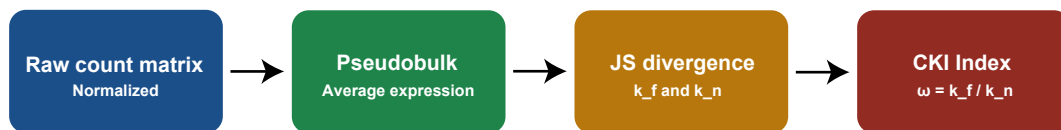
A



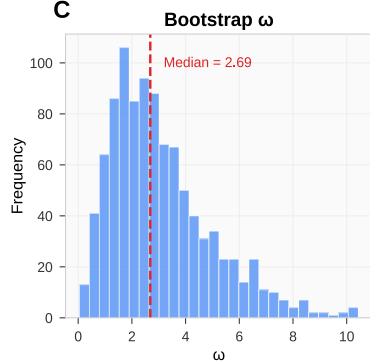
$$\omega = \frac{K_a}{K_s} \quad \omega > 1: \text{positive selection} \quad \omega \approx 1: \text{neutral drift} \quad \omega < 1: \text{purifying selection}$$

$$\text{CKI } \omega = \frac{k_f}{k_n} = \frac{JS(\text{softmax}(\epsilon_A[I]), \text{softmax}(\epsilon_B[I]))}{JS(\text{softmax}(\epsilon_A[H]), \text{softmax}(\epsilon_B[H]))} \quad \omega > 1: \text{selective transcriptomic remodeling}$$

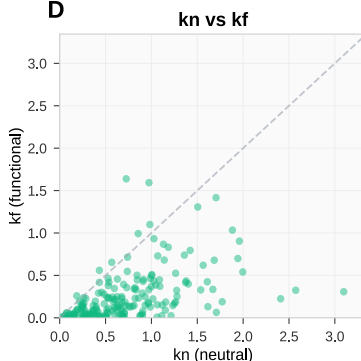
B



C



D



E

